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S11-07

Aim: Implement functions in Haskell

Assignment – 7(A)

Aim: Calculate NPR using Recursion

Code:

main.hs

```haskell
factorial:: Int->Int
factorial n = product [1..n]
npr:: Int->Int->Int
npr n r = div(factorial n)(factorial (n-r))
main :: IO()
main = do
    let n= 7
    let r= 3
    let answer = npr n r
    print("The answer for "++show n++"P"++show r++ " is:")
    print(answer)
```

Output:

```
Linking a.out ...
"The answer for 7P3 is:"
210


...Program finished with exit code 0
Press ENTER to exit console.
```

Assignment – 6(B)

Aim: Calculate Square root of of every number in input list

Code:

main.hs

```haskell
squareroot::[Double]-> [Double]
squareroot = map sqrt
main = do
putStrLn" The square root of the given list :"
print(squareroot[1,4,9,16,25])
print(squareroot[1,2,3,4,5])
```

Output:

```
 The square root of the given list :
[1.0,2.0,3.0,4.0,5.0]
[1.0,1.4142135623730951,1.7320508075688772,2.0,2.23606797749979]


...Program finished with exit code 0
Press ENTER to exit console.
```